

Nathan Tebbs

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SKILLS

Languages: C/C++, Java, Python, Go, Odin, JavaScript, TypeScript, Lua, Emacs Lisp, Bash, SQL, HTML/CSS

Tools & Technologies: Git, GitHub Actions, Docker, Linux, make, cmake, systemd, GDB, Valgrind, SSH, ufw, SQLite, PostgreSQL, VMware/KVM, Neovim & Emacs, Typst/LaTeX, Vercel, AWS, Google Cloud, Digital Ocean

Libraries: OpenGL, SDL3, Raylib, pandas, NumPy, scikit-learn, geopandas, statsmodels, Streamlit, React, Auth0, Clay, RGFW

EDUCATION

University of Arizona

Bachelor of Science in Computer Science

• Relevant Coursework: Compilers, Database Design, Algorithms, Computer Networking, Cloud Computing, Systems Programming

South Puget Sound Community College

Associate of Arts

Tucson, Arizona

August 2023 — May 2026

Olympia, Washington

September 2021 — May 2023

WORK EXPERIENCE

Contract Software Engineer

Sound Psychiatric Services

January 2021 — Present

Olympia, Washington

- Build and maintain the practice's public website, a landing page for around 100-120 patients that routes existing patients to their HIPAA-compliant payment and telehealth portals and informs prospective patients before an in-person or Zoom visit.
- Develop custom internal tools for the practice, including an application that reads and parses serial data from medical scale hardware to provide a live weight readout on a PC after the original display failed.
- Migrated hosting off a paid Squarespace business plan to a static setup, eliminating recurring hosting costs while preserving the existing email and domain in place so the transition caused no disruption. Built in plain HTML, CSS, and JavaScript to stay simple and low-maintenance.

Undergraduate Teaching Assistant (CSc 346 Cloud Computing)

University of Arizona

January 2025 — May 2025

Tucson, Arizona

- Assisted students with Linux-based cloud VMs, web servers, networking, and deployment issues; troubleshoot configuration errors and documented common failure modes and resolutions; worked with other teaching staff to meet course timelines on grading and office hours.

PROJECTS

Author, Mach: 2D Game Engine and Factory Game (github.com/vetr0s/mach)

June 2026 — Present

- Building a 2D game engine and the isometric factory game that runs on it, in pure C with no frameworks. The game is a routing puzzle: droppers spit out ore, conveyors carry it, upgraders raise its worth, and furnaces bank it as money you spend to expand the grid.
- Split the engine out into mach.h (github.com/vetr0s/mach.h), a zlib-licensed single-header library any C project can drop in and compile: an OpenGL 3.3 batch renderer with sprite atlases, a bitmap font, arena allocators, and a UI layout binding, with no mutable global state in engine code.
- Built hot reloading: the game compiles to a shared library the running host reloads on save, and because all state lives in host-owned memory, new code swaps in with no restart and no lost state. The simulation is fat structs in flat arrays out of one arena, built as a unity build in a single compiler call. CI tests on macOS, Linux, and Windows, and each git tag ships a static binary per platform.

Author, GateRelay: Secure Linux TCP Relay Service (github.com/vetr0s/gaterelay)

January 2026

- Built a TCP relay in Go forwarding public ingress traffic to a controlled backend, with connection limits, timeouts, graceful shutdown, and structured JSON logging.
- Hardened it with defense-in-depth: a non-root systemd user, sandboxing (capability dropping, syscall filtering, ProtectSystem), ufw firewall rules, and SSH key-only access, with a documented threat model.

Collaborator, Urban Crime Data Analysis (github.com/vetr0s/tucson-crime-analysis)

March 2025 — May 2025

- Ran a multivariate analysis of Tucson Police crime data against neighborhood income and streetlight locations to test how socioeconomic and environmental factors relate to crime.
- Built preprocessing, geospatial, and modeling pipelines in Python (pandas, geopandas, scikit-learn, statsmodels) using Ridge Regression, Random Forest, Logistic Regression, and OLS to predict high-crime wards.

EXTRACURRICULAR ACTIVITIES

FIRST Robotics Competition team 4450 (Safety Captain)

September 2021 — June 2023

- Led safety operations for a competitive engineering team, enforcing shop, electrical, and tooling safety standards under strict build and competition deadlines.
- Co-planned an international outreach program, coordinating risk mitigation for a \$60,000 fundraiser that sent 17 students and metal-based FTC robotics kits to Cairo, Egypt for hands-on robotics education.

Google Developer Student Club

August 2023 — May 2026

- Participated in peer-led workshops and collaborative projects on modern software development and Google Cloud technologies.

HackAZ 2026

April 2026 — April 2026

- Built and deployed GridWise Energy with 3 teammates in a 24-hour hackathon, a live web app analyzing energy usage from local power-grid data, on a React/TypeScript, Vercel, and Neon Postgres stack.
- Shipped auth, time-series usage tracking, AI chatbot insights, an ML forecast model, and an opt-in global leaderboard.